Solution to Word Scramble!

Anastasia Lee

1. After unscrambling: Law of Cosines
   We can see that 7 is the smallest number such that $\phi(n) = 6$, and then $\phi(8) = 4$, but $\phi(9) = 6$. The 9th letter is N.

2. After unscrambling: Radical Axis
   The smallest integer that has six divisors is 12, and $12_3 = 5$. The 5th letter is A.

3. After unscrambling: Hockey Stick Identity
   The only number that is the product of three numbers and the sum of those same three numbers is $6 = 1 \cdot 2 \cdot 3 = 1 + 2 + 3$. The 6th letter is S.

4. After unscrambling: Stewart's Theorem
   The average of the 15 consecutive numbers is $\frac{90}{15} = 6$, and this must be the middle number, meaning that the smallest number is -1. But then 0 is one of the numbers, so the entire product is 0. The 0th letter is S.

5. After unscrambling: Unit Vector
   We can see that -1 is a root, as $-1 - 2 + 7 - 4 = 0$, so $x + 1$ is a factor. Using polynomial long division or synthetic division, we get $x^3 - 2x^2 - 7x - 4 = (x + 1)(x^2 - 3x - 4)$. This can simply be factored as $(x + 1)(x + 1)(x - 4)$, so $x = 4$ is the positive root. The 4th letter is V.

6. After unscrambling: Euler's Identity
   We can see that none of the one-digit numbers will work, since $n < 3^n$ for all n. Moving on to two-digit numbers, $10 > 3^1 + 3^0$ and $11 > 3^1 + 3^1$, but $12 = 3^1 + 3^2$. The 12th letter is T.

7. After unscrambling: Cauchy-Schwarz Inequality
   The first perfect number is 6 $(1 + 2 + 3 = 6)$, and the second perfect number is 28 $(1 + 2 + 4 + 7 + 14 = 28)$. The 22nd letter is Y.

8. After unscrambling: Circumcircle
   We can express x recursively. If the first flip is heads, then we are done. If not, the expected number of flips to get heads is still x, but having used one flip already, the total will be $x + 1$. So, we get $x = \frac{1}{2} \cdot 1 + \frac{1}{2}(x + 1)$. Solving gives $x = 2$, so $2x = 4$. The 4th letter is U.

9. After unscrambling: Pythagorean Triple
   For a set of size n, there are 2^n possible subsets, since you can choose whether or not to include each element. So, the index is $2^{26}/2^{23} = 2^3 = 8$. The 8th letter is E.

10. After unscrambling: Angle Bisector
    The smallest potential answer is $2 \cdot 3 = 6$, but $3 - 2 = 1$, which is neither prime nor composite. The next smallest is $2 \cdot 5 = 10$, which works because $5 - 2 = 3$ and $5 + 2 = 7$ are both prime. The 10th letter is T.

Unscrambling the letters N, A, S, S, V, T, Y, U, E, T, we get STUYVESANT as our final answer.